

Security Incident Report

Incident Details

Name: Nitin Sukthe

Role: Security Analyst / SOC Analyst

Environment: Ubuntu Linux Lab Environment with Apache2 Web Server

Date: June 2026

Incident Title

Apache Web Server Log Analysis – Investigation of Suspicious HTTP Requests and 404 Error Activity

Section 1: Identify the Activity Observed During the Investigation

The activity observed during the investigation was web application reconnaissance against an Apache web server. Analysis of Apache access logs revealed multiple HTTP requests targeting non-existent resources including administrative and potentially sensitive directories such as:

```
/admin  
/login  
/test  
/secret
```

The requests generated HTTP 404 (Not Found) responses, indicating that the requested resources were unavailable on the server. These requests resemble behavior commonly associated with reconnaissance and directory enumeration activities performed by attackers attempting to discover administrative interfaces, hidden directories, or vulnerable application components.

The investigation also identified repeated HTTP GET requests to the root webpage ("/"), which represented normal user activity within the controlled lab environment. Apache error logs were reviewed and showed only normal service startup events without evidence of application failures, privilege escalation attempts, malware execution, or unauthorized configuration changes.

Based on the log evidence, the primary activity observed was web reconnaissance through directory and resource discovery attempts.

Section 2: Document the Incident Investigation

The investigation began after Apache web server logs were collected and reviewed in a controlled Ubuntu Linux environment. The objective was to analyze web traffic patterns, identify unusual requests, and determine whether any suspicious activity was present within the log data.

The security analyst first verified that the Apache service was operating normally and confirmed successful web server deployment through localhost testing. Once the service was confirmed operational, HTTP requests were generated to create activity within the Apache access logs.

The analyst navigated to the Apache log directory and reviewed both access and error logs. Access log analysis revealed successful HTTP requests to the root webpage and several failed requests resulting in HTTP 404 responses. The analyst used Linux command-line tools including `grep`, `awk`, `sort`, and `uniq` to filter log entries, identify the most active source IP address, and determine the most frequently requested resources.

Further investigation revealed requests targeting administrative and hidden resources including `"/admin"`, `"/login"`, `"/test"`, and `"/secret"`. These requests generated HTTP 404 responses because the resources did not exist on the server. In real-world environments, such activity is frequently associated with reconnaissance techniques used by attackers to identify potential attack surfaces.

The analyst then reviewed Apache error logs to identify any service failures, permission issues, or application errors. Error log analysis showed only standard Apache startup messages, indicating that the web server operated normally throughout testing.

The investigation concluded that no compromise occurred within the lab environment. However, the observed URL requests demonstrated behavior consistent with directory enumeration and web application reconnaissance techniques commonly observed during the early stages of cyber attacks.

Section 3: MITRE ATT&CK Mapping

The observed activity aligns with multiple MITRE ATT&CK techniques commonly associated with attacker reconnaissance and discovery operations.

T1595 – Active Scanning

The requests to `"/admin"`, `"/login"`, `"/test"`, and `"/secret"` resemble active scanning techniques used to identify accessible resources and application entry points.

T1083 – File and Directory Discovery

The attempted access to multiple directories is consistent with file and directory discovery behavior used to locate administrative portals, hidden content, and sensitive resources.

T1590 – Gather Victim Network Information

Repeated requests and resource discovery activities may support efforts to gather information regarding application structure and available services.

T1082 – System Information Discovery

Reconnaissance activities may ultimately contribute to identifying application technologies and system configurations.

Section 4: Security Findings

Finding 1: Multiple HTTP 404 Events

Evidence:

```
/admin  
/login  
/test  
/secret
```

Assessment:

The requests indicate attempted access to resources commonly targeted during reconnaissance activities.

Severity:

Low (Lab Environment)

Medium (Production Environment)

Finding 2: Source IP Analysis

Evidence:

```
127.0.0.1
```

Assessment:

All activity originated from localhost because the investigation was conducted within a controlled laboratory environment.

Severity:

Informational

Finding 3: No Critical Server Errors

Evidence:

```
AH00489: Apache configured -- resuming normal operations
AH00094: Command line: '/usr/sbin/apache2'
```

Assessment:

No service failures, application errors, or unauthorized modifications were identified.

Severity:

Informational

Section 5: Recommendations

One security measure recommended to improve detection and response capabilities is implementing centralized log monitoring through a Security Information and Event Management (SIEM) platform. SIEM solutions collect, correlate, and analyze log events from multiple systems, allowing analysts to identify suspicious patterns more effectively.

Additional recommendations include:

- Configure alerts for repeated HTTP 404 responses.
- Monitor requests targeting administrative resources.
- Deploy a Web Application Firewall (WAF).
- Restrict access to administrative interfaces.
- Review web server logs regularly.
- Correlate web server logs with authentication and firewall logs.
- Implement threat intelligence enrichment for suspicious source IP addresses.

These controls improve visibility into reconnaissance activity and strengthen an organization's ability to detect potential attacks before exploitation occurs.

Incident Conclusion

The Apache Web Server Log Analysis investigation successfully identified suspicious URL requests and HTTP 404 events consistent with reconnaissance and directory enumeration activity. Although no compromise occurred within the lab environment, the investigation demonstrated practical SOC analyst skills including log analysis, threat detection, event investigation, MITRE ATT&CK mapping, and security reporting.

The project highlights foundational blue-team capabilities used by Security Operations Centers to monitor web applications, identify suspicious behavior, and respond to potential threats.